

Facial Emotion recognition analysis using deep learning through RGB-D imagery of VR participants through partially occluded facial types

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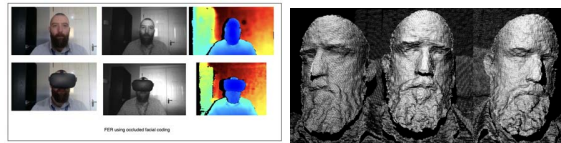


Figure 1: Left: Images taken to use in the FER algorithm to determine emotional state. Right: Truedepth camera raw RGB-D imagery

ABSTRACT

This research poster outlines the initial development of a facial emotion recognition (FER) evaluation system based on RGB-D imagery captured via a mobile based device. The study outlined features control group of non-occluded facial types and a set of participants wearing a head mounted display (HMD) in order to demonstrate an occluded facial type. We explore an architecture to develop a FER system that is suitable for occluded facial analysis. This paper details the methodology, experimental design and future work to be carried out to deliver such a system.

Index Terms: Human-centered computing—Visualization—; Human-centered computing—Collaborative and Social Computing—; Human-centered computing—Interaction Design—; Human-centered computing—Human Computer Interaction (HCI)—; Accessibility—Accessibility Technologies—; Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Facial Emotion Recognition (FER) is the process of facial analyses from RGB [4] and RGB-D imagery to categorise a subjects emotional state while engaging with a particular stimulus. As Virtual Reality (VR) technology has improved recently and a significant user base has been established the need for more immersive technologies to be developed to increase immersion and interactivity has increased. Within standard 2D experiences the entire face is used as a basis for detection and recognition of emotional state. However since a large proportion of the face is occluded while in VR recognition of the physiological state may not be tenable in this scenario. Due to this we propose the introduction of the depth component from RGB-D images in standard modern consumer devices in order to aid the training process for the recognition algorithm and to help alleviate lack of facial landmarks while wearing the HMD.

2 MOTIVATION

Monitoring of a VR users emotional state during a virtual experience has several advantages for the end user of the system. Across a spectrum of industries from monitoring educational engagement, analysing workplace satisfaction to interpreting a users sentiment towards a retail product or service. However as the global COVID-19

pandemic continues carrying out these experiments becomes more difficult as it requires in person analysis on premises in most circumstances. Our solution outlines a process in which the above scenarios are possible in an immersive and virtual world while still be able to perform facial coding for emotional state analysis. Furthermore we envisage this technology will aid in additional sectors as the use of face-masks and other head worn augmentations such as AR (augmented reality) smart glasses becomes more commonplace in consumer technology.

3 STAKEHOLDERS

This project will help to determine a VR user's emotional state while inside a VR experience. The goal of this study will be to evaluate emotion recognition algorithms trained with objective emotion datasets such as the DEAM [5] and DEAP [3]. Once a baseline accuracy has been established for the occluded face recognition we then will focus on improving the accuracy of recognition through modification of the weighting placed on specific feature points which will aid our stakeholders who are the HMD wearing userbase.

4 METHODOLOGY AND SYSTEM DESIGN

The following methodology was designed to carry out the study goals of this paper. Firstly the control group consisting of 20 participants will watch the objective emotion dataset video material while being monitored with the RGB-D video device, in this experiment Apple's front facing Truedepth camera on a 2021 Ipad Pro with LiDAR as seen in the system design 2 . A real time Wifi video stream from the ipad directly with appropriate timestamp requirements. This is achieved using record3D open source streaming software [1] which uses a real time streaming protocol to capture the live data required. This input will then be used with the CNN (convolutional neural network) based emotion recognition algorithm [2] to determine the current emotion state based on the valence and arousal model [6].

After this process we assign the original 20 participants a HMD to view the emotive dataset contents through. The same process is carried out again to gather the necessary data for an occluded facial image. This is again used as the source images for emotion recognition algorithm. The overall system design is tailored to be reproducible and repeatable with multiple weighting changes for the trained model. This allows our system to be semi-extensible to cater for differing facial feature coverings (such as facemasks).

5 EXPERIMENTAL DESIGN

A selection of participants to be contacted via randomised selection after a volunteer email will be sent out to staff within the institute. The participants are requested to firstly view the objective emotional data set contents in order to gauge a ground truth for that participants

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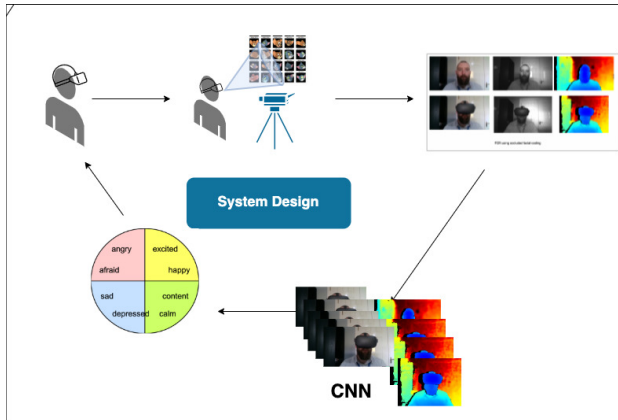


Figure 2: The System Design demonstrates the core components of the overall system, initial user data recording and demonstration of emotive contents, pre-processing of the captured RGB-D into their respective individual components, processing through the CCN based FER algorithm and finally the output displayed into the recognition accuracy score on the valence and arousal model

subjective emotional state. The RGBD once streamed is processed and a value returned for emotional state. The participant is then requested to wear the HMD and repeat the process in VR as seen in figure 3. A comparative analysis is then performed to determine concordance between the two sets of results. At this point we will be able to determine if any correlation can be made between the two sets of images.

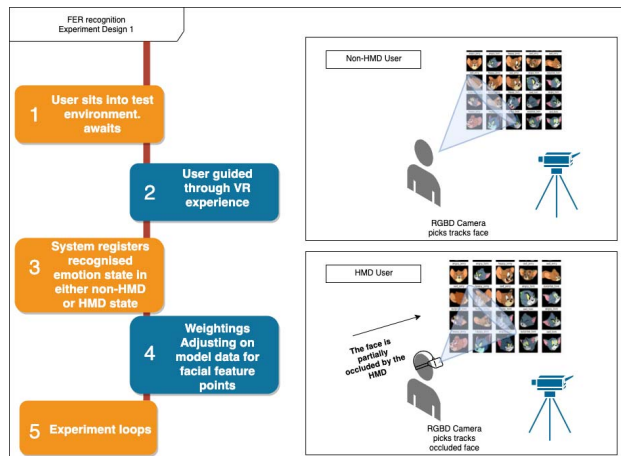


Figure 3: The Experimental Design of the study. It outlines the steps required from each participant of the study in order to gather required data both for occluded HMD users and non-HMD users alike. The data capture process through recording of the participant observing objective emotion contents.

This experiment can be replicated to determine if an increase in the accuracy of the recognition algorithm can be achieved. Using the same cohort of participants can allow the control group to give an indicative set of results against the standard accuracy result. Following on from the participant study with A/B testing to determine if the system performs as expected through the recognition accuracy score.

6 PRELIMINARY RESULTS AND FUTURE WORK

So far the system has been developed to capture and return live stream RGB-D data but as of yet we have yet to gather the number of participants from our volunteer group. We have been seeking to run the experiment remotely through participant's consumer phone devices and their personal HMDs. An alternative approach will be to use standard RGB-D dataset images, decimate the RGB-D images around the eyes and nasal bridge in order to replicate a completely occluded face but would raise issues with the depth component of the RGB-D image.

From our work to date we seek to develop a remote analysis tool that can be used in the users own home and expand the potential use cases of this technology. In future papers we would seek to alter the weightings for specific facial feature identification in order to more closely align recognition accuracy with full facial feature detection. More so we can alter the weighting to reflect other elements of occluded facial features such as those wearing a facemask or those with heavy facial hair. Furthermore as smart AR (Augmented reality) glasses become more common place we would be interested in performing a similar study to determine occlusion around AR headsets

7 CONCLUSION

Overall the results of the development to date are promising and open the door to a wider spectrum of research in future works. Initial development has adapted existing FER algorithms and future works will expand this concept to modify their weightings for partially occluded faces. We see great benefit in this study as the COVID-19 pandemic affects more and more everyday activities.

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